FRANCK JUNIOR ABOYA **MESSOU**

Passionate about Data Science, AI, and software development, I specialize in Python, Java, advanced algorithms, and big data analysis. Leveraging strong analytical skills and technical precision, I create efficient, scalable AI and software solutions tailored to solve complex challenges in IoT sensors, Big Data and modern applications.

https://mesabo.github.io

INFORMATIONS

+81 070 9037 4702
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Tokyo Koganei, Japan

LANGUAGES

Japanese: Beginner N5
English: Fluent
French: Native

SKILLS

Abilities

- Rigorous
- Analytic
- Adaptable

Skills

Programming Languages
- Python, Java, Dart, Bash, etc..

Artificial Intelligence
- Deep learning, LLM, IoT, Cloud dev

Databases
- SQL, NoSQL (MongoDB, Oracle)

Architecture & Dev
- APIs, Prototyping, Designing

INTERESTS

- Music
- Football
- Books

WORK EXPERIENCES

AI Research and Development

Academic Research (Tokyo) **Dec 2023 – Jul 2025**

- Developed Transformer-based algorithms to optimize Long-Term Energy Forecasting (paper) presented at **IEEE ICUFN 2025, July, Lisbon, Portugal**
- A **winner** of **IEEE VTS Tokyo/Japan Chapter 2024 Student Paper Award.**
- Development of RNNs algorithms for Energy Forecasting, resulting in a paper presented at **IEEE VTS2024-Fall, Washington DC.**

JICA Summer Internship -Japan AI Consulting, Osaka

- Worked with Generative AI, integrating Large Language Models (LLM) with backend API (Flask) and Database (log files + PostgreSql) using Cloud Dev.

FullStack Developer

Casys Technologies (Full-time, Ivory Coast) **Jan 2023 - Aug 2023**

- Improvement of backend systems in Java for commercial utility applications.
- Integrated backend systems with IoT devices such as vehicle sensors and door captors, enhancing system functionality.

BackEnd & FronEnd Developer

ANOHOU Solution (Part-time, Ivory Coast) **Au 2019 – Sep 2022**

- Developed systems , improving system performance and optimizing data management (Flutter, FastApi, NodeJs, GraphQL, MySQL, MongoDB).
- Member to the development of EKKOU app with ANOHOU Solutions, employing agile methodologies to deliver scalable solutions (Flutter)

PROJECTS

Research Assistance at *YU Lab* - Tokyo (2025 – 2026)

Advancing Optimization through Lightweight LLM for IoT-enabled smart grids.

Research Machine Learning and Data Analysis (2024 – 2025)

International Collaboration for Enhancing Transformer-based models for improved performance in Energy Management solutions.

EDUCATION

Ph.D DC1 in Applied Informatics

Sep 2025 – Sep 2028

Hosei University – Tokyo, Japan

Master 2 in Applied Informatics

Sep 2023 – Sep 2025

Hosei University – Tokyo, Japan

INTERNATIONAL COLLABORATIONS

- ***TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting with Learnable Positional Encodings and Attention Mechanisms,*** The 16th ICUFN 2025
- ***Federated Fine-Tuning of Large Language Models for Intelligent Automotive Systems with Low-Rank Adaptation,*** IEEE VTC2025-Spring.
- ***Dynamic IoT Resource Allocation Using Graph Reinforcement Learning,*** IEEE VTC2025-Spring.
- ***Enhancing Federated Learning in Consumer Electronics with Decoupled Knowledge Distillation against Data Poisoning,*** 2025 IEEE 22nd CCNC, doi: 10.1109/CCNC54725.2025.10976151.
- ***Enhancing Short-Term Load Forecasting in Internet of Things: A Hybrid Attention-based CNN-BiLSTM with Data Augmentation Approach,*** IEEE VTC2024-Fall.
- ***A Byzantine-Fault-Tolerant Federated Learning Method Using Tree-Decentralized Network and Knowledge Distillation for Internet of Vehicles,*** IEEE VTC2024-Fall.
- ***A Resource-efficient Text-to-Text Transfer Transformer Encoder-based Vertical Hybrid Model for Malicious URLs Detection,*** IEEE VTC2024-Fall.